



Ignoring the three-way interaction of digital orientation, Not-invented-here syndrome and employee's artificial intelligence awareness in digital innovation performance: A recipe for failure

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ABSTRACT

This study employs Knowledge-based view (KBV) and Transaction Cost Theory (TCT) to analyze the moderating effect of the Not-Invented-Here Syndrome (NIHS) on the relationship between digital orientation and digital innovation performance when employee's artificial intelligence awareness (EAIA) changes. EAIA alludes to employee's concern about being replaced by artificial intelligence at work. Structural equations were used to test the research model. The key findings reveal that when EAIA is high, the negative impact of the NIHS is stronger. The finding is significant because it confirms that the NIHS, which is associated with organizational rejection of external technologies, is not a phenomenon exclusive to the pre-digital era, but its negative consequences such as biases in the digital technology assessment and neglect or suboptimal use of them, persist in the digital era. Nevertheless, the major contribution demonstrates that the NIHS is not an isolated fact or a product of individuals' whims or irrationality; in the digital era, the NIHS is above all rational and opportunistic attitude rooted in EAIA. Furthermore, the NIHS becomes a planned response by employees aiming to minimize the risk of losing their jobs due to the impending replacement of digital technologies, particularly those related to automation processes.

1. Introduction

Digital technology is becoming more important in achieving business goals, and its pervasive effects have resulted in the radical restructuring of industries (Haefner et al., 2021). Digitization refers to how new digital technologies such as analytics, cloud, mobile, social media, or intelligent cyber-physical systems can be utilized to transform existing business processes (Arias-Pérez et al., 2021), and COVID-19 has unexpectedly turned out to be one of its main accelerators (Papagiannidis et al., 2020). Among all these technologies, the Deloitte firm recently discovered that companies are prioritizing investment in cloud, analytics, and process automation through intelligent cyber systems such as chatbots and virtual assistants, among others, to increase revenue and customer value, which can increase by 15% and 23% respectively in the early stage of digitization, and reach 45% and 41% respectively, when a digital maturity level is reached (Gurumurthy et al., 2020).

The analysis of the relationship between digital orientation and digital innovation performance is one of the emerging approaches in the literature; which has allowed scholars and practitioners to open the

black box of the relationship between digital transformation and specific features of organizational performance. Digital orientation is the deliberate strategic positioning of a firm to take advantage of the opportunities presented by those emerging digital technologies (Quinton et al., 2018). Thus, recent studies have shown that digital orientation significantly helps to improve digital innovation performance (Arias-Pérez et al., 2021; Liu et al., 2020), in terms of quantity, quality, distinction, novelty, and success, which refer to the superiority of digital solutions launched on the market or internally implemented in firms over competitors (Khin and Ho, 2019).

However, this emergent approach is a branch of traditional studies originating in the early 21st century on the generic relationship between technology orientation and innovation performance in the context of the pre-digital era (Narver et al., 2004). Individuals play the most important and central role because they have the knowledge to devise and implement new ways of creating value with emerging technologies (Sainio et al., 2012). Therefore, in this digital age, the situation is paradoxical because employees are needed by firms to understand the opportunities presented by digitization and specifically to generate digital

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innovations; for example, to help automate processes performed by some employees for years and which they know in detail, or to ensure that all new digital solutions can add value to customers when it is launched on the market (Frank et al., 2019; Haefner et al., 2021).

Nevertheless, employees are becoming increasingly aware of the high risk of losing their jobs when being replaced with new digital technologies (Nam, 2019); it is no longer a secret that digitization will result in a loss of more than 50% of jobs in the world (WEF, 2018) in all sectors of the economy (Agrawal et al., 2019), including high-tech and knowledge-intensive jobs (Dengler and Matthes, 2018). In this light, it is not surprising that individuals feel insecure, uncomfortable, undervalued, and unappreciated at work (Li et al., 2019; Nam, 2019). Thus, the situation is creating chaos for firms as these negative feelings trigger depression, indifference, and the desire to resign; and significantly decrease organizational commitment and career satisfaction (Brougham and Haar, 2018).

Some authors came up with the term “artificial intelligence and robotics awareness” to solve employees’ discomfort in the digital age (Arias-Pérez and Vélez-Jaramillo, 2021; Li et al., 2019), which specifically captures the extent to which employees view the likelihood of artificial intelligence and robotics awareness impacting their future career prospects negatively. Therefore, the notion alludes to the fact that employees perceive new digital technologies, particularly those linked with intelligent process automation, as a threat to their job/career (Brougham and Haar, 2018; Lingmont and Alexiou, 2020).

Unfortunately, the discussion on the new variable has mainly evolved at the individual’s level, while research has focused on mapping those negative feelings highlighted, and the decisions that individuals will likely make in a state of crisis (Li et al., 2019). Nevertheless, other authors have looked into identifying individual behaviors linked with the feeling of insecurity at work, hence, the personal use of certain technologies (Nam, 2019). Employees are still important in various organizational processes, including digital transformation, even when faced with the fear of losing their jobs. However, there is regrettably a lack of studies dealing with various organizational implications of employee’s artificial intelligence and robotics awareness.

Furthermore, Lichtenthaler (2019) recently proposed revisiting the discussion about the NIHS (Katz and Allen, 1982), albeit in the context of digital transformation, given his suspicions that this syndrome of internal resistance or organizational rejection of technologies coming from external sources (Nestle et al., 2019; Popa et al., 2017) could be jeopardizing the process of adoption of new digital technologies in firms. We believe that Lichtenthaler’s (2019) concern is reasonable, especially when employee’s artificial intelligence and robotics awareness is present. If this suspicion is correct, which is yet to be empirically proven, the NIHS would be a great inhibitor of digital transformation. In the pre-digital era, it has already been shown to alter and negatively affect how technology is sought and assessed in the environment (de Araújo-Burcharth et al., 2014), impede organizational learning (Hannen et al., 2019), and induce neglect or suboptimal use of external technology (Lichtenthaler and Ernst, 2006).

This study aims to analyze the moderating effect of the NIHS, moderated by employee’s artificial intelligence and robotics awareness, on the relationship between digital orientation and digital innovation performance. A moderated moderation model, better known as a three-way interaction, allows quantifying the rate of change of Xs effect (digital innovation) on Y (digital innovation performance) as the primary moderator changes when the secondary moderator varies (Hayes, 2017). In other words, the magnitude of the moderating effect of the NIHS would be determined by employee’s artificial intelligence and robotics awareness.

Thus, this study contributes to the discussion on barriers or inhibitors of digital transformation (Haefner et al., 2021; Horváth and Szabó, 2019). Firstly, it deals with the empirical analysis of the NIHS effects on the digitization processes proposed on a theoretical level. However, this syndrome is having devastating effects on digital transformation

processes, as internal resistance or organizational rejection of new digital technologies will weaken the relationship between digital orientation and digital innovation performance as described by Lichtenthaler (2019). Secondly, it slows down the search for digital technologies, distorts assessment, and induces the company to postpone its adoption. Thirdly, it delays digitalization, reducing the amount and pace in the development and launch of new digital solutions to the market. Hence, this study provides a better understanding of other aspects of the syndrome that have only been extensively analyzed in the context of the pre-digital era (Nestle et al., 2019; Popa et al., 2017).

Additionally, our study discovered that the syndrome will worsen if employee’s artificial intelligence and robotics awareness is present. Therefore, as people become more aware of the high chances of losing their jobs because of new digital technologies, their internal resistance or organizational rejection will increase as a strategy for protecting their interests, particularly as a measure to minimize the chances of losing their jobs. Hence, this study contributes to the basic discussion concerning the organizational implications of employee’s artificial intelligence and robotics awareness (Lingmont and Alexiou, 2020; Nam, 2019), particularly in the digital transformation process (Arias-Pérez and Vélez-Jaramillo, 2021).

The study also contributes to the discussion on digital transformation barriers. Thus, the most significant studies on the topic are exploratory and have been limited to identifying the most important barriers through qualitative studies (Horváth and Szabó, 2019; Stentoft et al., 2020). Nevertheless, there’s a lack of studies dealing with the analysis of the scope and magnitude of the barriers, i.e., to what extent they can be considered a real threat to digital transformation, particularly to the relationship between digital orientation and digital innovation performance. Surprisingly, this relationship has been previously analyzed under the assumption that the adoption of digital technologies automatically increases the number of new digital solutions without overcoming significant obstacles, that is, without recognizing the presence and impact of employees’ opportunistic behavior (Arias-Pérez et al., 2021; Khin and Ho, 2019; Liu et al., 2020). Hence, our study fills a gap in the literature by quantifying the negative impact of the two barriers represented in the moderated moderation, while questioning the perspective of previous studies that downplayed the influence of every type of negative organizational reaction in the relationship between digital orientation and digital innovation performance.

Furthermore, this study also contributes to the discussion on which theoretical lens could be the most appropriate to explain phenomena related to digital transformation. Therefore, recent studies have analyzed process automation processes through the lens of the behavioral theory of the firm based on the assumption that a firm may be understood as an algorithm or a combination of algorithms that process information (Haefner et al., 2021). Other studies have used a double lens: institutional theory and resource-based view theory to understand the adoption of certain digital technologies because of external pressures that produce changes to internal resources (Bag et al., 2021). However, due to variables in our three-way interaction, using the double lens of KBV of the firm and TCT is more appropriate. It has been previously used to analyze organizational phenomena where the tension between the need to create and share knowledge, and the need to protect personal interests is evident (Heiman and Nickerson, 2004; Sampson, 2004).

2. Theoretical framework and hypothesis statement

According to KBV, the inventory of individual and social knowledge is the most valuable resource of the organization (Grant, 1996) and the primary determinant of competitive advantage (Kogut and Zander, 1992). Knowledge is the combination of information that – by adding expert opinion, understanding, accumulated learning, or experience – yields valuable insights, know-how, and actionable instructions (Alavi and Leidner, 2001; Ardolino et al., 2018). According to this theoretical perspective, when the flow of knowledge in the firm is stirred up,

competitive advantages are generated. This perspective assumes that people, besides creating knowledge, are willing to share their idea so they can be used at all organizational levels (Martín-de Castro et al., 2011; Nonaka and Takeuchi, 1995).

Historically there has been a great debate between KBV and TCT, whose major assumption is opportunism; market players behave opportunistically, which is perceived as self-interest seeking with guile (Williamson, 1975). Opportunism is strongly linked to environmental uncertainty since it increases because of the unpredictability of an event – particularly market developments–, among other reasons (Williamson, 1996). Thus, firms are forced to structure themselves in a way in which controlling or reducing the threat of opportunism is facilitated. Consequently, the greatest criticism of KBV has been its belief that people act without precaution when sharing their knowledge (Heiman and Nickerson, 2004). TCT is hence the approach that has allowed us to recognize the existence of several counterproductive work behaviors that slow down the flow of knowledge in the firm (Arias-Pérez and Vélez-Jaramillo, 2021), such as the NIHS – deep-rooted in opportunism and the protection of personal interests (Antons and Piller, 2015; Nestle et al., 2019).

Accordingly, there has been a longstanding interest in the literature to reconcile both theoretical perspectives, KBV and TCT (Heiman and Nickerson, 2004); the understanding is that resorting simultaneously to the lens of both theories is required when analyzing organizational phenomena where the tension between the need to create and share knowledge and the need to protect personal interests is evident (Sampson, 2004). Therefore, KBV is essential for understanding the relationship between digital orientation and digital innovation performance, which is based on the inventory and flow of organizational knowledge. Conversely, TCT helps explain the interaction between the phenomena represented in moderated moderation, which assumes environmental uncertainty increases opportunistic behavior (Williamson, 1975, 1996). In our study, this uncertainty stems from the increasingly widespread use of new digital technologies in firms, particularly those related to intelligent process automation.

2.1. Digital orientation and digital innovation performance

The Strategic Orientation (SO) of an organization reflects its beliefs about conducting business and guides its activities (Gatignon and Xuereb, 1997; Zhou et al., 2005). Thus, there are different types of SO: customer orientation, competitor orientation, technology orientation, innovation orientation, among others (Theodosiou et al., 2012). For instance, competitor orientation assumes that the firm focuses on the permanent assessment of rivals' weaknesses and strengths in the short term; and on their capacities and strategy in the medium and long terms (Narver and Slater, 1990), which involves prioritizing tools such as benchmarking and business intelligence, to mention a few. SO toward digitalization or digital orientation has recently emerged as a branch of technology orientation, commonly understood as firms' intention to understand and use technology to respond to changes in the environment (Khin and Ho, 2019).

Digitization relates to how new digital technologies such as analytics, cloud, mobile, social media, or intelligent cyber-physical systems can be used to alter existing business processes (Verhoef et al., 2019; Vial, 2019). Digital orientation, by extension, is the deliberate strategic positioning of a firm to take advantage of the opportunities presented by those emerging digital technologies (Quinton et al., 2018), which involve the capture, transformation, and application of knowledge to create value from digital technologies (Arias-Pérez et al., 2021), whereas digital innovation performance refers to the superiority of digital solutions launched on the market or implemented internally in the firm in contrast with competitors, regarding quantity, quality, differentiation, novelty, and success (Khin and Ho, 2019; Liu et al., 2020).

Digital orientation is therefore crucial because it involves constant monitoring of new digital technologies and their various applications; it

encourages the creation of new ways to create value with these technologies, tailored to customers' needs, expectations, and anticipating future impacts (Burchardt and Maisch, 2019; Ismail et al., 2017). Thus, a climate that encourages experimentation with digital technologies is created in the firm, which ultimately allows firms to apply differentiated digital solutions quickly or add new digital components to the current offer of products and services (Vial, 2019).

For instance, it is becoming common in the local context for enterprises to have individuals or a digital transformation department who are tasked with participating in technology fairs where other sectors, firms, and technology providers present successful experiences and digital solutions that are generic or tailored to different sectors, like the financial sector, the retail sector, among others. Such department is also in charge of communicating new digitization trends to other areas of the organization and senior staff to devise, assess, acquire or develop digital solutions internally. The said organizational dynamic is allowing firms to venture into electronic payments; intelligent cyber systems, including chatbots, virtual assistants, recommendation agents to help customers make decisions –even stock trading bots and machine learning tools to segment or anticipate customer churn–; among other digital solutions. Therefore, the following hypothesis is put forward:

H1. Digital orientation positively influences digital innovation performance.

2.2. Moderating effect of the NIHS

The NIHS refers to internal resistance or rejection of technologies coming from external sources (Katz and Allen, 1982; Popa et al., 2017). The syndrome is manifested unconsciously or deliberately and can appear for several reasons, including individuals' need to define, express and defend their self-identity, self-concepts, values, and to maintain their social relationships (Arias-Pérez et al., 2017). The second reason is the confirmation bias, which is only placing importance on external digital technologies that do not deviate from individuals' opinions and points of view about the use of digital technologies (Hannan et al., 2019). Third, the syndrome has a utilitarian function related to the opportunism of individuals, specifically their interest in obtaining benefits and avoiding situations that jeopardize their interest (Antons and Piller, 2015).

Digital transformation success lies largely in the swiftness with which new digital technologies are detected and adopted (Cassetta et al., 2020; Ferreira et al., 2019). Hence, the syndrome is a significant inhibitor of digitization, particularly of the positive effect of digital orientation on digital innovation performance, as shown to skew and negatively affect how technology is sought and assessed in the environment (de Araújo-Burchard et al., 2014; Popa et al., 2017); additionally, it slows down organizational learning (Hannan et al., 2019) and induces neglect or suboptimal use of external technology (Lichtenthaler and Ernst, 2006).

Consequently, concerning the syndrome, first, it slows down the search for digital technologies, biases their assessment, and induces the company to postpone its adoption (Lichtenthaler, 2019). In other words, it delays digitalization, reducing the amount and pace in the development and launch of new digital solutions to the market. For instance, some emerging firms have belatedly started to utilize chatbots because they find it difficult to abandon their preference for personal interaction and believe this approach is the only way to improve customer experience. Second, it entails a slow learning process of digital technologies and their various applications, which implies a reluctance to grasp the opportunities they offer for the business in terms of enhancing customer experience and operational efficiency. Thus, it hinders the ability to speculate on ways of creating value from digital technologies. Third, it implies a suboptimal use of digital technologies, limiting the utilization of their benefits and the possibility of generating digital solutions with a high degree of quality difference from the competition in terms of good technical performance and the novelty of their functionalities for the

market.

Additionally, there are other well-publicized cases in which the relationship between digital orientation and digital innovation performance is hindered by the syndrome. For example, in the financial sector, efforts are being made to integrate computer applications from several technology providers, which are designed in various programming languages, to heighten and enhance customers' experience with digital financial services. However, firms have generalized opinions about which technology provider offers the most comprehensive computer system to support customers' credit card transactions, what the best Customer Relationship Management on the market is; the same occurs with the various applications. Consequently, due to these biases, firms are reluctant about testing and migrating to new applications which, despite offering more generic functionalities, better integrate into the firm's IT systems. This includes the two stated challenges above, making it difficult to launch or implement in the firm digital solutions superior to the competition and tailored to customer needs and expectations. Therefore, the following hypothesis is proposed:

H2. The impact of digital orientation on digital innovation performance will be reduced when the NIHS is present.

2.3. Moderating effect of employee's artificial intelligence and robotics awareness

Artificial intelligence and robotics awareness captures the extent to which employees view the likelihood of artificial intelligence and robotics awareness impacting their future career prospects negatively (Li et al., 2019; Lingmont and Alexiou, 2020). Thus, as the digital transformation in the sector or firm, deepens, employees begin perceiving new digital technologies as a threat to their jobs/careers, especially those associated with intelligent process automation, (Arias-Pérez and Vélez-Jaramillo, 2021; Brougham and Haar, 2018). What happens in this instance is that internal resistance or organizational rejection of new digital technologies is accentuated as employees' strategy to protect their interests, and minimize the risk of losing their jobs (Lichtenthaler, 2019). Employee's artificial intelligence and robotics awareness intensifies the utilitarian function of the NIHS, where employees act opportunistically to avoid the situation that threatens their interest

(Antons and Piller, 2015).

Hence, employee's artificial intelligence and robotics awareness intensifies the negative consequences of the NIHS, particularly the bias in the search and assessment of digital technologies in the environment (de Araújo-Burcharth et al., 2014); hinders organizational learning (Hannen et al., 2019); and induces neglect or suboptimal use of external technology (Lichtenthaler and Ernst, 2006). For example, there are suspicions that individuals are sabotaging digital transformation: employees deliberately fail to fully share or avoid disclosing crucial and detailed information to the engineers responsible for automation, namely the step-by-step of the organizational process or the task under their charge in the firm. Such behavior prevents the success of the digital solutions; hence, causing a high likelihood of intelligent robots not adding value, generating negative experiences for internal and external customers (Canhoto and Clear, 2020). Therefore, the following hypothesis is posed:

H3. The negative impact of the NIHS on the relationship between digital orientation and digital innovation performance will be greater when employee's artificial intelligence and robotics awareness is present.

3. Methodology

3.1. Sample and data collection

The proposed model (see Fig. 1) was tested with a sample of a high, medium, and low-technology manufacturing firms (Eurostat, 2009), and service firms (see Table 1), from sectors in which the adoption of intelligent robots is on the rise (Butner and Ho, 2019; Ransbotham et al., 2017). These firms are located in Colombia, an emerging country classified as technology follower (Castellacci, 2011), and given this two-fold condition, firms that have been successful in the pre-digital economy, but which have been forced to accelerate their digital transformation processes to remain competitive, predominate in the country (Arias-Pérez et al., 2021). Consequently, in the South American continent, Colombian firms, along with their Brazilian counterparts, are investing the most in new digital technologies, even more than Chile and Argentina (Portulans Institute, 2019).

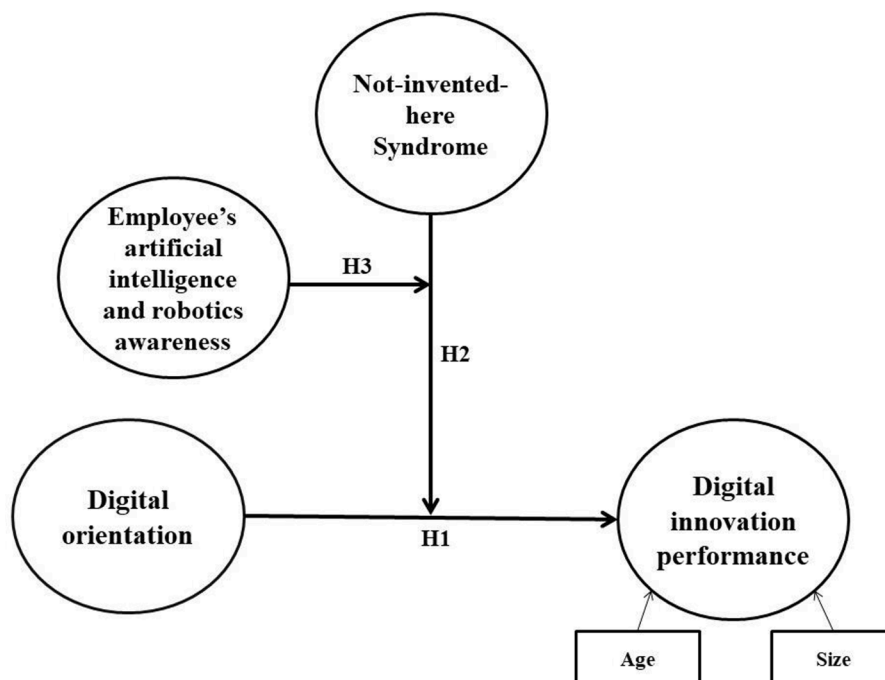


Fig. 1. Research model.

Table 1
Sample characterization.

Sector	Activity	Frequency	%
Manufacturing	Manufacture of rubber and plastic products	6	8%
	Other low and medium technology manufacturing sectors	19	24%
	High technology manufacturing sectors	6	8%
Services	Financial and insurance services	12	15%
	Telecommunications	8	10%
	Wholesale and retail trade	6	8%
	Education	5	6%
	Architectural and engineering activities	4	5%
	Other less knowledge-intensive service	8	10%
	Other knowledge-intensive services	5	6%
Size (Number of employees)			
	Large (≥ 250)	44	56%
	SMEs (≥ 10 – < 250).	35	44%
Respondent's position			
	Systems and Technology	29	37%
	Presidency or General Management	15	19%
	Research and Development	10	13%
	Finances	6	
	Marketing	5	6%
	Human resources	2	3%
	Other areas	23	15%

Fieldwork was conducted in the summer of 2020 using an electronic questionnaire emailed to the management staff with a total of 130 firms served by an emerging company that offers them digital transformation services, mainly in automation processes. In detail, the firms in the sample require two main types of services: first, IT solutions for all things relating to networks; infrastructure to enable private, public (Microsoft Azure and Amazon Web Services) or multi- cloud services; and computer security and connectivity, namely network optimization and software-defined networking. Second, the surveyed companies require Robotic Process Automation (RPA) solutions through the UiPath platform. RPA services regularly on demand include computer vision, reengineering, and process design solutions for their automation; licensing of the quantity and type of intelligent robots (attended or unattended) required for executing these processes; and a tool to schedule and monitor this digital workforce. However, there is also a demand for data analytics solutions. These include the implementation of analytical models based on unstructured data (NoSQL) in the private or public cloud, either previously developed or failing that, developed from scratch with a team of data scientists; for example, to predict work disabilities; segment and anticipate churn; or predict propensity to buy.

In our study, we used the minimum R-squared method to estimate a minimum sample size allowing us to reach an acceptable level of statistical power (80%) to guarantee that the statistical test performed was able to recognize a path coefficient as statistically significant ($P < 0.05$) (Hair et al., 2019; Kock and Hadaya, 2018). Finally, 79 valid responses were obtained –one manager per company–, a sample size guaranteeing a satisfactory statistical power above 80% (Hair et al., 2017).

3.2. Measurement scales

To measure digital orientation, the scale of Arias-Pérez et al. (2021) was adapted. The scales of Khin and Ho (2019) and Sok and O'Cass (2011) were adapted to measure digital innovation performance. In turn, the scale of Araújo-Burcharth et al. (2014) was used for the syndrome, while for employee's artificial intelligence and robotics awareness we adapted the scale of Brougham and Haar (2018) (see Appendix A). Besides, a Likert scale going from *very inferior to competitors* (1) to *Very superior to competitors* (5) was used for digital innovation performance. For the remaining constructs, a different Likert scale going from *totally disagree* (1) to *totally agree* (5) was employed.

The study included two control variables that have previously proven their influence on innovation performance in the pre-digital era:

age and size (Yu and Lee, 2017); they have also been used with this purpose in recent empirical studies on digital innovation performance (Arias-Pérez et al., 2021). Size corresponds to the natural logarithm of the number of employees, while age corresponds to the natural logarithm of the years since the creation of the business.

3.3. Common method variance

Regarding the strategies for avoiding the Common Method Variance (CMV) problem, we adopted Harman's single-factor test (Podsakoff and Organ, 1986), which allowed us to establish that the variance accounted for in the first factor is 47.08%. Therefore, these results indicate that the likelihood of the CMV problem occurring in the data is very low (Kock, 2021).

3.4. Reliability and validity

The reliability and validity of the measurement model were examined with equations through the Partial Least Squares (PLS) method (Hair et al., 2019). As for individual reliability, it was verified that all items had a loading equal to or above 0.7, or close to 0.6 in the case of scales used occasionally in the literature (Hair et al., 2017) in the small number of empirical works, such as the case of the NIHS. It was also verified that all constructs had a Cronbach's alpha, Composite Reliability (CR) indices, and Average Variance Extracted (AVE) greater than 0.5 (Benitez et al., 2020). Table 2

3.5. Discriminant validity

To establish discriminant validity, meeting the Fornell-Larcker criterion was initially verified; particularly, Table 3 shows the square root of the AVE exceeds the correlations between the different constructs. Secondly, it was confirmed that all values of the Heterotrait-Monotrait (HTMT) correlations were below the threshold of 0.85 (Hair et al., 2019).

3.6. Moderating test procedure

A moderated moderation model, also known as a three-way

Table 2
Reliability and validity.

Constructs	Loadings	AVE	CR	CA
Digital orientation		0.70	0.94	0.93
Dig_Orient1	0.81*			
Dig_Orient2	0.84*			
Dig_Orient3	0.86*			
Dig_Orient4	0.80*			
Dig_Orient5	0.80*			
Dig_Orient6	0.85*			
Dig_Orient7	0.86*			
Digital innovation performance		0.78	0.96	0.95
Dig_Innoperfom1	0.92*			
Dig_Innoperfom2	0.92*			
Dig_Innoperfom3	0.80*			
Dig_Innoperfom4	0.81*			
Dig_Innoperfom5	0.91*			
Dig_Innoperfom6	0.93*			
Dig_Innoperfom7	0.87*			
Employee's artificial intelligence and robotics awareness		0.86	0.95	0.93
IA_Awareness1	0.94*			
IA_Awareness2	0.92*			
IA_Awareness3	0.92*			
Not-invented-here syndrome		0.52	0.76	0.64
NIH1	0.94*			
NIH2	0.59*			
NIH3	0.58*			

* $p < 0.001$.

Table 3

Discriminant validity.

Constructs	Mean	Standard deviation	Fornell-Larcker				HTMT		
			1	2	3	4	1	2	3
1. Digital innovation performance	3.39	1.10	0.88						
2. Digital orientation	3.67	1.11	0.77	0.83			0.81		
3. Employee's artificial intelligence and robotics awareness	2.25	1.16	0.13	0.22	0.93		0.13	0.21	
4. Not-invented-here syndrome	2.33	1.22	−0.22	−0.26	0.06	0.72	0.21	0.33	0.24

AVE square root on the diagonal in bold; correlations between constructs below the diagonal.

interaction, allows quantifying the rate of change of X's effect (digital innovation) on Y (digital innovation performance) as the primary moderator (NIHS) changes when the secondary moderator (employee's artificial intelligence and robotics awareness) varies (Hayes, 2017).

To test the moderating effect, two structural equations models were estimated. In the first, the control variables, age and size, were introduced. In the second the moderating variable was incorporated: NIHS, moderated by employee's artificial intelligence and robotics awareness. Structural equations were used through the PLS method, which allows obtaining the confidence intervals at 95% and the t values of the coefficients of the different paths from a bootstrapping of 5000 subsamples (Hair et al., 2019). For the moderating effect analysis, the orthogonalizing approach was used since it eliminates the issue of collinearity through residual centering. Additionally, it has superiority in terms of parameter and prediction accuracy (Ali-Memon et al., 2019).

4. Results

Models 1 and 2 show that the control variables of age and size do not have any influence on digital innovation performance (see Table 4). In contrast, the positive and significant effect of digital orientation on innovation performance both in Model 1 ($\beta = 0.75$) and Model 2 ($\beta = 0.72$) is noticeable; therefore, H1 is accepted. Additionally, it is observed in Model 2 that both moderating variables have a negative and significant effect, both the NIHS ($\beta = -0.17$) and employee's artificial intelligence and robotics awareness ($\beta = -0.20$). H2 and H3 are thus accepted.

However, the f^2 value of 0.23 indicates the moderating effect size of the NIHS moderated by employee's artificial intelligence and robotics awareness is found above the low (0.02) and medium (0.15) threshold level, but far from the high level (0.35) (Hair et al., 2017). Fig. 2 shows to what extent the syndrome modifies the relationship between digital orientation and digital innovation performance, and to what extent employee's artificial intelligence and robotics awareness intensifies the syndrome's negative effects.

Table 4

Structural model results.

Constructs	Model 1	Model 2	Hypothesis
Control variables			
Age	−0.11	−0.07	
Size	0.01	0.07	
Main effects			
Digital orientation	0.75**	0.72**	H1
Not-invented-here syndrome (NIHS)		0.05	
Employee's artificial intelligence awareness (EAIA)		−0.04	
Interaction effects			
Digital orientation*NIHS		−0.17*	H2
Digital orientation*EAIA		0.12	
NIHS*EAIA		−0.03	
Digital orientation*NIHS*EAIA		−0.20*	H3
F^2		0.23	
R^2	0.61	0.68	
Q^2	0.46	0.51	

* $p < 0.05$; ** $p < 0.001$.

Additionally, regarding the evaluation of our research model's predictive power, Table 4 shows that Model 2 explains 68% of the digital innovation performance variance, which indicates its predictive power is above the moderate level (0.50) and close to the substantial one (0.75). This result is quite representative of a study like ours which analyzes phenomena scarcely studied in the literature since it evidences there is a solid theoretical development behind our research model (Benitez et al., 2020). However, the R^2 only indicates the model's in-sample explanatory power – it says nothing about the model's out-of-sample predictive power (Hair and Sarstedt, 2021). Therefore, we proceeded to calculate the Q^2 value, which is an indicator combining aspects of out-of-sample prediction and in-sample explanatory power. This value is calculated through the blindfolding procedure, which removes single points in the data matrix, imputes the removed points with the mean, and estimates the model parameters (Hair et al., 2019). Table 4 shows that the Q^2 value of Model 2 is 0.51, indicating its predictive accuracy exceeds the threshold of the large level (0.50) and is well above the medium level (0.25).

Lastly, we strictly evaluated the out-of-sample predictive power through PLSpredict, a method that uses training and holdout samples to generate and evaluate predictions from PLS path model estimations (Shmueli et al., 2016). PLSpredict first randomly splits the total sample into 10 subgroups equal in size. The method then selects nine subgroups, which, in combination, become the training sample. The remaining subgroup becomes the holdout sample that the training sample attempts to predict. The process of predicting a subgroup rotates through all of the original 10 subgroups so that each of the 10 subgroups is used by itself as a holdout sample, and the other 9 subgroups are combined to predict that holdout sample. When this process is completed, the data for every respondent in the total sample have been predicted by a training sample that did not include that respondent's data to estimate the model results (Hair et al., 2020). Table 5 shows the error prediction values of our PLS structural equation model (Model 2): Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) are inferior compared with the values of a Linear Regression (LR) model, which means our research model has high out-of-sample power (Hair et al., 2019).

5. Discussion

Our results confirm that not everything is perfect in digital transformation processes, whose risk of failure is high if the influence of the NIHS and employee's artificial intelligence and robotics awareness is ignored. Regarding the syndrome, as expected, it is present when devising and implementing new ways to create value with digital technologies. In the digital age, the syndrome retains some of its traditional characteristics, such as the confirmation of bias, thus, giving relevance only to external technologies that do not contradict organizational beliefs concerning the use of digital technologies. This trend is usually a regular manifestation of the syndrome in firms located in less technologically advanced countries (Arias-Pérez et al., 2017), where acquiring or licensing, and parameterizing digital technologies take precedence over developing them internally from scratch.

Nevertheless, the findings confirm our greatest concern about the syndrome in the digital age: employee's artificial intelligence and

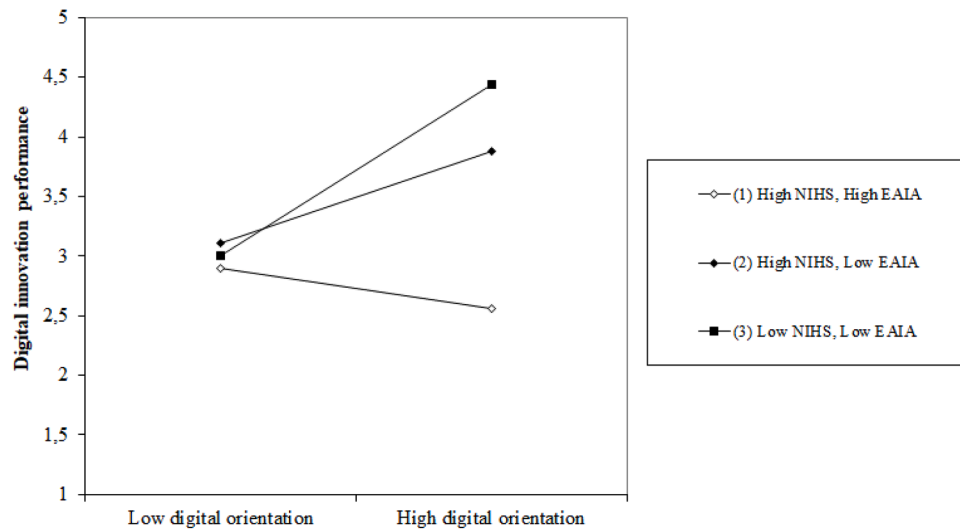


Fig. 2. The three-way interaction of digital orientation, not-invented-here syndrome, and employee's artificial intelligence awareness in digital innovation performance.

Table 5
PLSpredict.

Independent construct items	PLS RMSE	MAE	MAPE	LR RMSE	MAE	MAPE
Dig_Innoperfom1	0.82	0.62	24.00	331.07	42.07	2925.43
Dig_Innoperfom2	0.88	0.70	26.39	402.44	49.31	3406.86
Dig_Innoperfom3	0.90	0.64	27.74	268.75	35.47	2465.01
Dig_Innoperfom4	0.81	0.61	23.52	282.66	35.86	2447.96
Dig_Innoperfom5	0.70	0.55	19.92	356.51	45.65	3174.54
Dig_Innoperfom6	0.73	0.57	22.00	265.64	32.78	2322.84
Dig_Innoperfom7	0.73	0.53	20.66	146.52	21.32	676.34

robotics awareness intensifies its most worrisome function: the utilitarian function, in which internal resistance or organizational rejection of new digital technologies seeks to avoid negative outcomes at all costs—in this case being replaced by an intelligent machine. In other words, our findings demonstrate that in the digital transformation process, neither common sense nor individuals' emotions are to be underestimated. Employees are aware of the great threat that automation brings along; therefore, they can easily go from being the main allies in this process to being its staunchest saboteurs as their interests are seriously threatened with the arrival of digital technologies.

However, our results only manage to uncover the first layer of an organizational reality that can be much more complex and challenging for digital transformation processes. The NIHS remains a lesser problem as much more severe things are happening beneath; for instance, deliberate and aggressive concealment of knowledge in all its forms (Arias-Pérez and Vélez-Jaramillo, 2021). Even more serious, knowledge sabotage instances may be occurring in addition to the syndrome (Serenko and Choo, 2020). This means that to safeguard their interests, employees can act maliciously and be determined to reprimand, humiliate, and neutralize other employees who are decisively supporting the digital transformation processes.

In sum, our results indicate that the syndrome constrains the positive effect of digital orientation on digital innovation performance. Specifically, the syndrome slows down the search for digital technologies, biases their assessment, and induces the company to delay its adoption. Consequently, the syndrome reduces the possibility to timely develop and launch new digital solutions in the market. It obstructs the ability to think of new methods of creating value from digital technology and to take advantage of the opportunities they provide to improve customer experience and operational efficiency. However, the syndrome implies a suboptimal use of digital technologies, thereby limiting the possibility of

generating digital solutions with a high degree of distinction for competitors, regarding quality and good technical performance, and the novelty of their functionalities for the market. When EAIA is high, these negative consequences of the syndrome, of an already considerable magnitude, as indicated by the f^2 value of 0.23, are intensified when employees' artificial intelligence awareness is high.

6. Conclusions

In terms of academic contributions, the study enriches the discussion on digital transformation inhibitors or barriers (Haefner et al., 2021; Horváth and Szabó, 2019). As such, our results empirically confirm the suspicions of Lichtenthaler (2019) in the sense that the NIHS is sabotaging digitization processes. Thus, there is indeed an internal resistance or organizational rejection of digital technologies that are generating adverse effects on firms, weakening the relationship between digital orientation and digital innovation performance. This finding is significant because it confirms that the NIHS is not a phenomenon exclusive to the pre-digital era as it has been widely conceived in the literature, and particularly in recent empirical studies (Nestle et al., 2019; Popa et al., 2017). In the digital age, its negative consequences remain with regard to biases in the search and assessment of digital technologies in the external environment, as well as their neglect, suboptimal use, or exploitation.

However, the most significant theoretical contribution of the study is showing that the NIHS is neither an isolated event in the digital age nor a product of the whims and irrationality of individuals, as was argued in the past (Arias-Pérez et al., 2017; Hannen et al., 2019). In contrast, internal resistance or organizational rejection of digital technologies is primarily a rational attitude rooted in employee's artificial intelligence and robotics awareness. From a different perspective, the utilitarian function of the syndrome is predominant in the digital age because it ends up becoming a deliberate reaction of employees seeking to minimize the risk of losing their jobs in order not to be replaced by digital technologies, including those associated with process automation, such as intelligent cyber-physical systems and their several applications—e.g., chatbots, virtual assistants, recommendation agents, stock trading bots—, among others. In sum, our results constitute a turn in the way the syndrome has been analyzed, which has dismissed its irrational nature in the context of digital transformation.

Thus, our study contributes to redefining the notion of employee's artificial intelligence and robotics awareness, by highlighting a nuance, specifically its role as a trigger and intensifier of opportunistic

organizational behaviors, such as the NIHS. This situation goes in contrast with the theoretical perspective of previous studies that focused either on analyzing their direct impact on variables that fall within the individual domain, as the intention to quit (Li et al., 2019; Lingmont and Alexiou, 2020), or identifying individual behaviors that are related to a lesser feeling of job instability (Nam, 2019). In detail, our research offers a different study perspective that identifies the serious consequences caused by employee's artificial intelligence and robotics awareness in organizational processes; in the case of digital innovation, workers continue to be important adding their knowledge and experience. However, both aspects remain crucial for devising and implementing new ways of creating value with digital technologies (Frank et al., 2019; Haefner et al., 2021). To summarize, our study constitutes a turning point regarding the study of EAIA, showing that its organizational consequences deserve more attention than they have had in the field of digital transformation, considering the excessive research interest on their repercussions for the individual.

Furthermore, the study also contributes to the discussion on digital transformation barriers. Firstly, it raises awareness of new barriers, which complement the lists recently created through qualitative studies (Horváth and Szabó, 2019; Stentoft et al., 2020). In contrast to these exploratory investigations, our study succeeds in quantifying concretely the negative impact of the two barriers represented in the moderated moderation. Simultaneously, it contributes to positioning this type of analysis as a promising methodological alternative for advancement in the field of study, considering the need to establish to what extent the barriers are a threat in specific digital transformation situations such as the relationship between digital orientation and digital innovation performance. In particular, our results indicate that the syndrome slows down the search for digital technologies, biases their assessment, and induces the company to postpone its adoption. In other words, the syndrome delays digitalization, reducing the amount and pace of the development and launch of new digital solutions to the market. Therefore, this result also forces a reconsideration of the assumption that has guided the study concerning the relationship between digital orientation and digital innovation performance, according to which, the adoption of digital technologies automatically increases the number of new digital solutions without encountering significant obstacles or recognizing the presence and impact of adverse organizational reactions (Arias-Pérez et al., 2021; Khin and Ho, 2019; Liu et al., 2020).

However, the high predictive power of our research model indicates our double theoretical lens based on KBV and TCT is appropriate for the analysis of organizational phenomena where there is an evident tension between the need for dynamizing the flow of organizational knowledge and the need to protect personal interests, in the context of digital transformation. Thus, our study contributes to expanding the spectrum of theoretical perspectives in this emerging field of study, which is related to extending the research horizon beyond the possibilities and explanatory limits of the most widely used theories to date – behavioral theory of the firm, institutional theory, and the resource-based view theory. Hence, the combination of KBV and TCT offers the possibility of asking different research questions about digital transformation, bringing to the forefront phenomena with negative connotations which have barely been studied so far in the literature. This study also reinforces the idea of re-evaluating the convenience of using a single theoretical approach when the complexity of digital transformation requires researchers to repeatedly adopt a double lens (Bag et al., 2021).

Therefore, this study puts a spin on the study of the relationship between digital orientation and digital innovation performance in emerging countries, since our results give room to question the approach followed by previous studies that have completely ignored the presence of opportunistic organizational behaviors (Arias-Pérez et al., 2021; Khin and Ho, 2019; Liu et al., 2020). Unlike what happens in developed countries where many firms can realize digital transformation based on internal resources and capabilities (Geissinger et al., 2020), the opposite happens in emerging countries that are becoming advanced

technologically. Neglecting the presence of this type of behavior is thus a major research error because firms' digitization in these countries largely depends on the acquisition of external technological solutions. Besides, in the pre-digital era – it was clear that behaviors of this type, such as the NIHS, are more worrisome and harmful in emerging countries (Arias-Pérez et al., 2017; Bogers et al., 2019) than in developed countries (Greco et al., 2019). Consequently, another considerable academic contribution of our study is that it demonstrates the importance of this perspective that emphasizes the study of these types of inhibitors, which may short-circuit the flow of digital technology knowledge from external sources, essential in emerging countries.

Regarding the practical implications of the study, the consensus in the literature is that training programs have been recognized as a way to alleviate both the NIHS (Hannen et al., 2019) and the negative effects of employee's artificial intelligence and robotics awareness (Li et al., 2019). In other words, they are means of reducing employees' resistance to innovation processes or for individuals to acquire new skills that allow them to take on the tasks that digital technologies cannot perform, either full-time or part-time. Training would create opportunities for employees to be reassigned to other positions. However, we believe the major goal is to promote a culture of transparency in the firm in which employees know in advance the risks they face because of intelligent automation. Simultaneously, it should be combined with outplacement strategies and an aggressive system of economic incentives that encourage voluntary retirement, in addition to motivating individuals to support documentation and, above all, automation of the processes they have been in charge of so long and which they know in detail.

Conversely, although the number of respondents ($n = 79$) is acceptable and the statistical technique used requires that a bootstrapping of 5000 subsamples be run (Hair et al., 2019), the sample size can be considered as a limitation. However, this limitation contrasts with other notable methodological strengths of the work. First, despite its size, the sample is well-suited for the study's purpose, given that the companies surveyed are deeply involved in digital transformation processes. However, the measurement model shows excellent reliability and validity indicators, highlighting not just the quality of the measurement scales but also that of the data collected. Nonetheless, the main methodological strength of our study is undoubtedly the high predictive power, which is representative of work like ours that analyzes phenomena scarcely studied in the literature. It shows there is a solid theoretical development behind our research model (Benitez et al., 2020; Hair and Sarstedt, 2021).

However, the main limitation concerns the fact that our results are limited to companies from emerging countries that are becoming advanced technologically (Castellacci, 2011) and, although been successful in the pre-digital economy, have been pushed to venture into digitization to remain competitive. In this sense, the digitization strategy focuses on the acquisition of digital solutions so they can be adapted or parameterized according to organizational requirements instead of prioritizing their developments with internal resources and capabilities (Arias-Pérez et al., 2021).

Nevertheless, generalizing our results to born-digital companies such as Uber and Airbnb (Geissinger et al., 2020) would encounter limitations, since developing digital solutions with their in-house team of engineers is prioritized by their digitization strategy. Consequently, in this context, the NIHS could be evidence through other nuances, to confirm their self-identity, self-concepts, and values when solutions created by competitors or other external entities are assessed (Nestle et al., 2019; Popa et al., 2017). Therefore, the utilitarian function of the syndrome might be irrelevant in this context. Apparently, under these conditions, there is little room for situations related to employee's artificial intelligence and robotics awareness to arise due to the high percentage of processes that are automated in these types of firms from the moment they emerge.

Therefore, future research should focus on analyzing the NIHS in born-digital firms to examine in more detail what syndrome features are

evidenced when digital innovations are being created. It would also make sense to check for any trace of employee's artificial intelligence and robotics awareness. However, the focus of future studies should be on in-depth analysis of organizational reaction toward the adoption of new digital technologies beyond the NIHS, using a broader sample of pre-digital firms. Based on our findings, we suspect bigger challenges concerning more complex situations for digital transformation. It would then be sensible to examine whether employee's artificial intelligence and robotics awareness triggers certain forms of counterproductive knowledge behaviors such as knowledge sabotage (Arias-Pérez and Vélez-Jaramillo, 2021; Serenko and Choo, 2020).

Hence, it would be pertinent to study failed cases in terms of digital transformation; the literature is already warning in that aspect, in some circumstances, automation is destroying value because it creates bad experiences for the customer (Canhoto and Clear, 2020). We suspect that in defending their interests, employees may go beyond internal resistance or rejection of new digital technologies: they could even be committing serious acts of knowledge sabotage such as providing erroneous information related to the tasks for which they are responsible (Serenko and Choo, 2020) to create limitation process automation and cause it to fail. Therefore, it would be appropriate for future studies to fully open this black box of organizational reaction to digitization.

Therefore, we also deem that it is necessary to analyze the inhibitors of digital transformation from another perspective. It will be relevant for

advancing the field of study to examine the extent to which classical drivers of innovation can help neutralize the negative effects of the moderated moderation analyzed in our study; for example, transformational leadership (Singh et al., 2020) and innovative culture (Santos-Vijande et al., 2021), which have been proven to be variables that help to keep innovation intact, even in times of great uncertainty resulting from technological change. However, we consider that in a context where firms will inevitably strengthen intelligent process automation to remain competitive, future studies should expeditiously focus on determining the extent to which variables linked to employees' trust in management intentions, such as organizational justice and management trustworthiness (Cui and Jiao, 2019), can reduce the negative moderating effect of the NIHS and alter EAIA.

CRedit authorship contribution statement

José Arias-Pérez: Conceptualization, Writing – review & editing, Funding acquisition. **Juan Vélez-Jaramillo:** Writing – original draft, Formal analysis, Visualization.

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Appendix A. Scale items

Constructs	Items
Digital orientation	
Dig_Orient1	Developing a clear vision regarding how new digital technologies (Social media, Mobile, Analytics, Cloud Computing) contributes to business value
Dig_Orient2	Integrating business strategy and digital strategy
Dig_Orient3	Enabling functional area and general management's ability to understand value of new digital technology investments
Dig_Orient4	We constantly keep current with new digital technology innovations
Dig_Orient5	We are able to and continue to experiment with new digital technologies as necessary
Dig_Orient6	We have a climate that is supportive of trying out new ways of using digital technologies
Dig_Orient7	We constantly seek new ways to enhance the effectiveness of digital technology use
Digital innovation performance	
Dig_Innoperfom1	The number of digital solutions introduced in the market is superior compared to our competitors'
Dig_Innoperfom2	The number of successful digital solutions is superior compared to our competitors'
Dig_Innoperfom3	The time to market for digital solutions is superior compared to our competitors'
Dig_Innoperfom4	The quality of our digital solutions is superior compared to our competitors'
Dig_Innoperfom5	The features of our digital solutions are superior compared to our competitors'
Dig_Innoperfom6	The applications of our digital solutions are totally different from our competitors'
Dig_Innoperfom7	Some of our digital solutions are new to the market at the time of launching
Employee's artificial intelligence and robotics awareness	
IA_Awareness1	Our employees are worried because the work they currently do could be done by robots or artificial intelligence
IA_Awareness2	Our employees are worried about their future in the company because they could be replaced by machines or artificial intelligence
IA_Awareness3	Our employees are worried about their future in the industry because they could be replaced by robots or artificial intelligence
Not-invented-here syndrome	
NIH1	Our employees have negative attitudes to applying ideas and technologies from outside
NIH2	Our employees regard the application of external knowledge as valuable as the application of knowledge generated inside
NIH3	Our employees are positive toward developing new ideas, solutions and technologies for other companies*

*Reverse code

References

- Agrawal, A., Gans, J.S., Goldfarb, A., 2019. Artificial intelligence, the journal of economic perspectives. *Am. Econ. Assoc.* 33 (2), 31–50.
- Alavi, M., Leidner, D.E., 2001. Knowledge management and knowledge management systems: conceptual foundations and research issues. *MIS Q., JSTOR* 25 (1), 107–136.
- Ali-Memon, M., Cheah, J.-H., Hiram-Ting, T.R., Chuah, F., Huei-Cham, T., 2019. Moderation analysis: issues and guidelines. *J. Appl. Struct. Equ. Model.* 3 (1), i–xi.
- Antons, D., Piller, F.T., 2015. Opening the black box of 'not invented here': attitudes, decision biases, and behavioral consequences. *Acad. Manag. Perspect.* 29 (2), 193–217.
- de Araújo-Burcharth, A.LdA., Knudsen, M.P., Søndergaard, H.A., 2014. Neither invented nor shared here: the impact and management of attitudes for the adoption of open innovation practices. *Technovation* 34 (3), 149–161.
- Ardolino, M., Rapaccini, M., Saccani, N., Gaiardelli, P., Crespi, G., Ruggeri, C., 2018. The role of digital technologies for the service transformation of industrial companies. *Int. J. Prod. Res.* 56 (6), 2116–2132.
- Arias-Pérez, J., Perdomo-Charry, G., Castaño-Ríos, C., 2017. Not-invented-here syndrome and innovation performance: the confounding effect of innovation

- capabilities as organisational routines in service firms. *Int. J. Innov. Manag.* 21 (1), 1750036.
- Arias-Pérez, J., Vélez-Jaramillo, J., 2021. Understanding knowledge hiding under technological turbulence caused by artificial intelligence and robotics". ahead-of-print. *J. Knowl. Manag.* ahead-of (ahead-of) <https://doi.org/10.1108/JKM-01-2021-0058>.
- Arias-Pérez, J., Vélez-Ocampo, J., Cepeda-Cardona, J., 2021. Strategic orientation toward digitalization to improve innovation capability: why knowledge acquisition and exploitation through external embeddedness matter? *J. Knowl. Manag.* 25 (5), 1319–1335. <https://doi.org/10.1108/JKM-03-2020-0231> ahead-of-p No. ahead-of-print.
- Bag, S., Pretorius, J.H.C., Gupta, S., Dwivedi, Y.K., 2021. Role of institutional pressures and resources in the adoption of big data analytics powered artificial intelligence, sustainable manufacturing practices and circular economy capabilities. *Technol. Forecast. Soc. Change* 163, 120420.
- Benitez, J., Henseler, J., Castillo, A., Schubert, F., 2020. How to perform and report an impactful analysis using partial least squares: guidelines for confirmatory and explanatory IS research. *Inf. Manag.* 57 (2), 103168.
- Bogers, M., Burcharth, A., Chesbrough, H.W., 2019. Open innovation in Brazil: exploring opportunities and challenges. *Int. J. Innov.* 7 (2), 178–191.
- Brougham, D., Haar, J., 2018. Smart technology, artificial intelligence, robotics, and algorithms (STARA): employees' perceptions of our future workplace. *J. Manag. Organ.* 24 (2), 239–257.
- Burchardt, C., Maisch, B., 2019. Digitalization needs a cultural change – examples of applying agility and open innovation to drive the digital transformation. *Procedia CIRP* 84, 112–117.
- Butner, K., Ho, G., 2019. How the human-machine interchange will transform business operations. *Strategy Leadersh.* 47 (2), 25–33.
- Canhoto, A.I., Clear, F., 2020. Artificial intelligence and machine learning as business tools: a framework for diagnosing value destruction potential. *Bus. Horiz.* 63 (2), 183–193.
- Cassetta, E., Monarca, U., Dileo, I., Di Berardino, C., Pini, M., 2020. The relationship between digital technologies and internationalisation. Evidence from Italian SMEs. *Ind. Innov.* 27 (4), 311–339.
- Castellacci, F., 2011. Closing the technology gap? *Rev. Dev. Econ.* 15 (1), 180–197.
- Cui, Y., Jiao, H., 2019. Organizational justice and management trustworthiness during organizational Change: interactions of Benevolence, Integrity, and managerial approaches. *Inf. Process Manag.* 56 (4), 1526–1542.
- Dengler, K., Matthes, B., 2018. The impacts of digital transformation on the labour market: substitution potentials of occupations in Germany. *Technol. Forecast. Soc. Change* 137, 304–316.
- Eurostat, 2009. 'High-technology' and 'knowledge based services' aggregations based on. NACE Rev. 2. Eurostat, Luxembourg.
- Ferreira, J.J.M., Fernandes, C.I., Ferreira, F.A.F., 2019. To be or not to be digital, that is the question: firm innovation and performance. *J. Bus. Res.* 101 (November), 583–590.
- Frank, A.G., Mendes, G.H.S., Ayala, N.F., Ghezzi, A., 2019. Servitization and Industry 4.0 convergence in the digital transformation of product firms: a business model innovation perspective. *Technol. Forecast. Soc. Change* 141, 341–351.
- Gatignon, H., Xuereb, J.-M., 1997. Strategic orientation of the firm and new product performance. *J. Mark. Res.* 34 (1), 77–90.
- Geissinger, A., Laurell, C., Sandström, C., 2020. Digital disruption beyond Uber and Airbnb—tracking the long tail of the sharing economy. *Technol. Forecast. Soc. Change* 155, 119323.
- Grant, R.M., 1996. Toward a knowledge-based theory of the firm. *Strateg. Manag. J.* 17 (S2), 109–122. John Wiley & Sons, Ltd.
- Greco, M., Grimaldi, M., Cricelli, L., 2019. Benefits and costs of open innovation: the BeCO framework. *Technol. Anal. Strateg. Manag.* 31 (1), 53–66.
- Gurumurthy, R., Schatsky, D., Camhi, J., 2020. Uncovering the Connection Between Digital Maturity and Financial Performance. Deloitte Insights, p. 23.
- Haefner, N., Wincent, J., Parida, V., Gassmann, O., 2021. Artificial intelligence and innovation management: a review, framework, and research agenda. *Technol. Forecast. Soc. Change* 162, 120392.
- Hair, J.F., Howard, M.C., Nitzl, C., 2020. Assessing measurement model quality in PLS-SEM using confirmatory composite analysis. *J. Bus. Res.* 109, 101–110.
- Hair, J.F., Risher, J.J., Sarstedt, M., Ringle, C.M., 2019. When to use and how to report the results of PLS-SEM. *Eur. Bus. Rev.* 31 (1), 2–24.
- Hair, J.F., Hult, G.T.M., Ringle, C., Sarstedt, M., 2017. A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM). SAGE, Los Angeles.
- Hair, J.F., Sarstedt, M., 2021. Explanation plus prediction—The logical focus of project management research. *Proj. Manag. J.* 52 (4), 319–322. SAGE Publications Inc, p. 8756972821999945.
- Hannen, J., Antons, D., Piller, F., Salge, T.O., Coltman, T., Devinney, T.M., 2019. Containing the Not-Invented-Here Syndrome in external knowledge absorption and open innovation: the role of indirect countermeasures. *Res. Policy* 48 (9), 103822.
- Hayes, A.F., 2017. Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach. Guilford Publications, New York, NY, USA.
- Heiman, B.A., Nickerson, J.A., 2004. Empirical evidence regarding the tension between knowledge sharing and knowledge expropriation in collaborations. *Manag. Decis. Econ.* 25 (67), 401–420.
- Horváth, D., Szabó, R.Z., 2019. Driving forces and barriers of Industry 4.0: do multinational and small and medium-sized companies have equal opportunities? *Technol. Forecast. Soc. Change* 146, 119–132.
- Ismail, M.H., Khater, M., Zaki, M., 2017. Digital Business transformation and strategy: what do we know so far? *Manufacturer Article* (November 2017), 36.
- Katz, R., Allen, T.J., 1982. Investigating the Not Invented Here (NIH) syndrome: a look at the performance, tenure, and communication patterns of 50 R & D Project Groups. *R D Manag.* 12 (1), 7–20.
- Khin, S., Ho, T.C., CF and International, H.T., 2019. Digital technology, digital capability and organizational performance: a mediating role of digital innovation. *Int. J. Innov. Sci.* 11 (2), 177–195 edited by.
- Kock, N., 2021. Harman's single factor test in PLS-SEM: Checking for common method bias. *Data Analysis Perspectives Journal* 2 (2), 1–6.
- Kock, N., Hadaya, P., 2018. Minimum sample size estimation in PLS-SEM: the inverse square root and gamma-exponential methods. *Inf. Syst. J.* 28 (1), 227–261. John Wiley & Sons, Ltd.
- Kogut, B., Zander, U., 1992. Knowledge of the firm, combinative capabilities, and the replication of technology. *INFORMS* 3 (3), 383–397.
- Li, J., Bonn, M.A., Ye, B.H., 2019. Hotel employee's artificial intelligence and robotics awareness and its impact on turnover intention: the moderating roles of perceived organizational support and competitive psychological climate. *Tour. Manag.* 73, 172–181.
- Lichtenthaler, U., 2019. Extremes of acceptance: employee attitudes toward artificial intelligence. *J. Bus. Strategy* 41 (5), 39–45.
- Lichtenthaler, U., Ernst, H., 2006. Attitudes to externally organising knowledge management tasks: a review, reconsideration and extension of the NIH syndrome. *R D Manag.* 36 (4), 367–386. John Wiley & Sons, Ltd.
- Lingmont, D.N.J., Alexiou, A., 2020. The contingent effect of job automating technology awareness on perceived job insecurity: exploring the moderating role of organizational culture. *Technol. Forecast. Soc. Change* 161 (October 2019), 120302.
- Liu, J., Chang, H., Forrest, J.Y.-L., Yang, B., 2020. Influence of artificial intelligence on technological innovation: evidence from the panel data of china's manufacturing sectors. *Technol. Forecast. Soc. Change* 158, 120142.
- Martín-de Castro, G., López-Sáez, P., Delgado-Verde, M., 2011. Towards a knowledge-based view of firm innovation. Theory and empirical research. *Journal of Knowledge Management* 15 (6), 871–874. <https://doi.org/10.1108/13673271111179253>.
- Nam, T., 2019. Technology usage, expected job sustainability, and perceived job insecurity. *Technol. Forecast. Soc. Change* 138, 155–165.
- Narver, J.C., Slater, S.F., 1990. The effect of a market orientation on business profitability. *J. Mark.* 54 (4), 20–35.
- Narver, J.C., Slater, S.F., MacLachlan, D.L., 2004. Responsive and proactive market orientation and new-product success. *J. Prod. Innov. Manag.* 21 (5), 334–347.
- Nestle, V., Täube, F.A., Heidenreich, S., Bogers, M., 2019. Establishing open innovation culture in cluster initiatives: the role of trust and information asymmetry. *Technol. Forecast. Soc. Change* 146, 563–572.
- Nonaka, I., Takeuchi, H., 1995. The Knowledge-Creating Company: How Japanese Companies Create the Dynamics of Innovation. Oxford University Press, New York, NY, USA.
- Papagiannidis, S., Harris, J., Morton, D., 2020. WHO led the digital transformation of your company? A reflection of IT related challenges during the pandemic. *Int. J. Inf. Manag.* 55, 102166.
- Podsakoff, P.M., Organ, D.W., 1986. Self-reports in organizational research: problems and prospects. *J. Manag.* 12 (4), 531–544.
- Popa, S., Soto-Acosta, P., Martínez-Conesa, I., 2017. Antecedents, moderators, and outcomes of innovation climate and open innovation: an empirical study in SMEs. *Technol. Forecast. Soc. Change* 118 (1), 134–142.
- Portulans Institute, 2019. Network Readiness Index 2019. Washington, DC, USA.
- Quinton, S., Canhoto, A., Molinillo, S., Pera, R., Budhathoki, T., 2018. Conceptualising a digital orientation: antecedents of supporting SME performance in the digital economy. *J. Strateg. Manag.* 26 (5), 427–439.
- Ransbotham, S., Kiron, D., Gerbert, P., Reeves, M., 2017. Reshaping business with artificial intelligence: closing the gap between ambition and action. *MIT Sloan Manag. Rev.* 59 (1), 1–17.
- Sainio, L.-M., Ritala, P., Hurmelinna-Laukkanen, P., 2012. Constituents of radical innovation—Exploring the role of strategic orientations and market uncertainty. *Technovation* 32 (11), 591–599.
- Sampson, R.C., 2004. Organizational choice in R&D alliances: knowledge-based and transaction cost perspectives. *Manag. Decis. Econ.* 25 (67), 421–436.
- Santos-Vijande, M.L., López-Sánchez, J.A., Pascual-Fernández, P., Rudd, J.M., 2021. Service innovation management in a modern economy: insights on the interplay between firms' innovative culture and project-level success factors. *Technol. Forecast. Soc. Change* 165, 120562.
- Serenko, A., Choo, C.W., 2020. Knowledge sabotage as an extreme form of counterproductive knowledge behavior: the role of narcissism, Machiavellianism, psychopathy, and competitiveness. *J. Knowl. Manag.* 24 (9), 2299–2325.
- Shmueli, G., Ray, S., Velasquez Estrada, J.M., Chatla, S.B., 2016. The elephant in the room: predictive performance of PLS models. *J. Bus. Res.* 69 (10), 4552–4564.
- Singh, S.K., Del Giudice, M.D., Chierici, R., Graziano, D., 2020. Green innovation and environmental performance: the role of green transformational leadership and green human resource management. *Technol. Forecast. Soc. Change* 150, 119762.
- Sok, P., O'Cass, A., 2011. Achieving superior innovation-based performance outcomes in SMEs through innovation resource-capability complementarity. *Ind. Mark. Manag.* 40 (8), 1285–1293.
- Stentoft, J., Aadsbøll Wickstrøm, K., Philipsen, K., Haug, A., 2020. Drivers and barriers for Industry 4.0 readiness and practice: empirical evidence from small and medium-sized manufacturers. *Production Planning and Control*. Taylor & Francis, pp. 1–18.
- Theodosiou, M., Kehagias, J., Katsikea, E., 2012. Strategic orientations, marketing capabilities and firm performance: an empirical investigation in the context of frontline managers in service organizations. *Ind. Mark. Manag.* 41 (7), 1058–1070.

- Verhoef, P.C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi, D., J., Fabian, N. and Haenlein, M. (2019), "Digital transformation: a multidisciplinary reflection and research agenda", *J. Bus. Res.*, available at: 10.1016/j.jbusres.2019.09.022.
- Vial, G., 2019. Understanding digital transformation: a review and a research agenda. *J. Strateg. Inf. Syst.* 28 (2), 118–144.
- WEF (2018), *The Future of Jobs Report 2018*, World Economic Forum, Geneva.
- Williamson, O.E., 1975. *Markets and Hierarchies: Analysis and Antitrust Implications: A Study in the Economics of Internal Organization*. Free Press, New York, NY, USA.
- Williamson, O.E., 1996. *The Mechanisms of Governance*. Oxford University Press, New York, NY, USA.
- Yu, G.J., Lee, J., 2017. When should a firm collaborate with research organizations for innovation performance? The moderating role of innovation orientation, size, and age. *J. Technol. Transf.* 42 (6), 1451–1465.
- Zhou, K.Z., Gao, G.Y., Yang, Z., Zhou, N., 2005. Developing strategic orientation in China: antecedents and consequences of market and innovation orientations. *J. Bus. Res.* 58 (8), 1049–1058.
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